

ORIGINAL PAPER

Adjacent Satellites Collaborative Beam Hopping–Based
Interference Avoidance Technique for LEO Satellite SystemChao Zhang^{1,2}  | Xiangyu Gan^{1,2} | Guang Liang^{1,2,3} | Xinglong Jiang^{1,2,3} | Huawang Li^{1,2,3}¹Innovation Academy for Microsatellites, Chinese Academy of Sciences, Shanghai, China | ²University of Chinese Academy of Sciences, Beijing, China | ³Shanghai Engineering Center for Microsatellites, Shanghai, China**Correspondence:** Guang Liang (liangg@microsat.com)**Received:** 2 August 2025 | **Revised:** 16 December 2025 | **Accepted:** 6 January 2026**Keywords:** collaborative beam hopping | greedy user scheduling | GSO–LEO coexistence | power allocation | resource efficiency (RE)**ABSTRACT**

Low earth orbit (LEO) satellite communication system can achieve global coverage and has attracted widespread attention. However, the common demand for broadband satellite spectrum resources inevitably leads to cofrequency interference between LEO and geostationary orbit (GSO) satellite communication systems. To meet interference avoidance requirements while ensuring LEO system performance, this paper proposes an adjacent satellites collaborative beam hopping–based interference avoidance technique for GSO–LEO coexistence systems. First, the adaptive demand satisfaction ratio user grouping method is developed to accommodate the energy efficiency weight factor in resource efficiency (RE). Meanwhile, a residual demand priority adjacent satellites collaborative greedy user scheduling scheme is employed to enhance LEO system capacity while avoiding severe interference to GSO systems. Subsequently, a quadratic transformation–based convex optimization power allocation is proposed to further improve LEO system RE and reduce cofrequency interference. Finally, the above steps are alternately iterated timeslot by timeslot to obtain the local optimal user scheduling strategy and power allocation scheme over the beam-hopping period. Simulation results validate that the proposed technique effectively mitigates interference while achieving optimal performance in multiple indicators including RE.

1 | Introduction

Satellite communication systems, characterized by their wide area coverage, support widespread application in diverse fields, including navigation, broadcasting, the Internet of Things, and military communications [1]. Compared with satellite communication system operating in other orbits, the low earth orbit (LEO) satellite communication system has recently been regarded as the most promising for future development due to the unique advantages of low latency, reduced transmission loss, and lower operational costs [2, 3]. Many LEO communication constellations have been proposed and gradually deployed around the world, typical ones such as SpaceX, OneWeb, and Telesat [4, 5]. However, the available frequency resources of satellites are limited, which directly affects the

communication capacity and performance of the satellite systems. In accordance with the prevailing frequency allocation guidelines of the International Telecommunication Union (ITU), the broadband LEO satellite communication systems primarily employ Ku/Ka band for information transmission, which offer relatively wider bandwidth. These frequency bands coincidentally overlap with those commonly used by geostationary orbit (GSO) broadband satellite systems [6]. The dynamic coverage properties of LEO satellites may lead to overlapping coverage areas with GSO systems, which results in the intersystem cofrequency interference [7]. In compliance with ITU frequency compatibility requirements, non-geostationary orbit (NGSO) satellite systems, including LEO constellations, must implement certain interference avoidance measures to prevent themselves from causing harmful

interference to GSO satellite systems [8]. Therefore, effective interference avoidance techniques are the prerequisite and foundation for the operation of broadband LEO communication constellation systems.

Numerous interference avoidance schemes have been proposed in the existing literature for cofrequency interference mitigation between LEO and GSO satellite systems. Spatial isolation is a widely adopted interference avoidance measure to reduce interference by restricting the angle between the communication links of different systems, including GSO arc avoidance and exclusion zone [9, 10], but it usually brings communication interruption and service blind spots. Zhang and Jiang [11] conducted simulation analysis based on this technique to determine the minimum satellite count necessary for ensuring continuous communication services while also meeting interference constraints. Cognitive radio effectively mitigates intersystem interference by detecting spectrum holes in the GSO satellite system before accessing them, but it requires a balance between spectrum sensing accuracy and spectrum utilization efficiency [12]. Gu et al. [13] optimized communication parameters for both satellite systems based on spectrum sharing principles, but this approach requires additional modifications to the GSO satellite networks, which does not comply with current ITU intersystem frequency compatibility regulations. Du et al. [14] applied interference alignment technology to compress the interference signal space at the LEO satellite transmitter to decrease interference to the GSO satellite system, but this method requires accurate channel state information (CSI). Considering the challenge of obtaining perfect instantaneous CSI in the high mobility communication scenarios of LEO satellites, Jalali et al. [15] and Zheng et al. [16] proposed a beamforming scheme based on average CSI and a RIS-aided beamforming scheme based on statistical CSI, respectively, to mitigate intersystem interference. However, both average and statistical CSI fail to match the rapid channel variations, which limits interference suppression performance of these beamforming schemes. Moreover, the RIS-aided beamforming requires the deployment of additional equipment, increasing system complexity.

The existing interference mitigation techniques reviewed above decrease the interference experienced by the GSO satellite system originating from the LEO satellite system to some extent. However, these approaches do not account for the system characteristics of operational LEO networks. Given the traffic exhibits an uneven distribution across temporal and spatial dimensions, beam-hopping technology remains a prevalent method for providing services in LEO broadband satellite systems, both currently and in the foreseeable future [17, 18]. Consequently, there is an urgent need to develop a frequency compatibility scheme tailored for beam-hopping LEO satellite systems and GSO satellite systems. Moreover, constrained by factors such as size and weight, satellite communication systems are inherently resource limited, with onboard energy almost entirely derived from solar panels [19, 20]. Unlike traditional beam-hopping systems that focus solely on system capacity, this paper adopts resource efficiency (RE) [21], a widely used metric in 5G/6G green communications, as the key performance indicator. Specifically, this paper aims to maximize RE under the constraint of effectively mitigating interference to GSO satellite systems, which presents a challenging problem.

To effectively address the above problem, we propose adjacent satellites collaborative beam-hopping interference avoidance scheme, which maximizes system RE adhering to interference avoidance requirements by sharing beam-hopping resource between adjacent satellites. Specifically, the complex resource optimization problem, incorporating timeslots and power allocation under interference avoidance constraints, is decomposed into multiple suboptimization problems based on timeslot sequencing to reduce algorithmic complexity. Each suboptimization problem within a timeslot consists of two optimization tasks—user scheduling and power allocation—solved using a greedy algorithm and quadratic transformation, respectively. As the objective function is updated per timeslot, the proposed scheme effectively achieves a local optimum of the original problem. Simulation results indicate that the proposed algorithm effectively mitigates interference to GSO satellite systems while surpassing other methods in multiple indicators, including demand satisfaction rate, system capacity, and RE. This paper's primary contributions are outlined as follows.

- In order to achieve frequency compatibility between LEO and GSO satellite systems, this paper formulates a multi-satellite collaborative beam-hopping resource allocation problem under interference avoidance constraints, jointly optimizing user scheduling timeslots and satellite power allocation to enhance system RE.
- The NP-hard nature of the original problem poses significant challenges to directly addressing the above problem. This paper proposes an adaptive demand satisfaction ratio grouping and residual demand priority greedy timeslot allocation strategy, alongside a convex optimization power allocation scheme based on quadratic transformation, to address user scheduling and beam power allocation within each timeslot, respectively. By iteratively updating the objective function per timeslot, the proposed approach effectively approximates a local optimum of the original problem.

The rest of this paper is organized as follows. The system model is introduced in Section 2, while the multisatellite collaborative beam-hopping resource allocation problem is formulated. Section 3 provides the details of the algorithm designed to address this problem. In Section 4, simulation results show the effectiveness of the proposed method. Finally, this paper is concluded in Section 5.

2 | System Model and Problem Formulation

2.1 | System Model

In this paper, we consider a representative downlink spectrum coexistence scenario between common Walker star constellation and GSO satellite system, as illustrated in Figure 1. Each LEO satellite serves cells within a certain range from its sub-satellite point in considered case. To simplify the analysis, the cells within the coverage area are assumed to be arranged in a hexagonal cellular structure. The Walker star constellation is selected because it is a typical constellation with low deployment cost among the global coverage constellation, widely adopted and representative. Satellites in different orbital

applicable to all orbital planes. Moreover, when evaluating the interference from the LEO satellite system to GSO users, all visible satellites from the entire LEO constellation across multiple orbital planes are taken into account.

Considering the coverage range of GSO satellites and the locations of cells potentially affected by interference, this paper assumes that M LEO satellites serve a total of N cells. The cells that LEO satellites can serve are constrained by elevation angle; typically, only N_h cells with high-elevation angle can be covered, as indicated by the solid green lines representing the coverage range in Figure 1. However, this inevitably results in severe interference from LEO satellites to GSO satellite users in certain cells, particularly when LEO satellites are located at low latitudes, where collinear interference, the most detrimental cofrequency interference, is highly likely to occur. To address this, the proposed method considers that the service area of each LEO satellite includes both its own high-elevation-angle cells and those of adjacent satellites, as denoted by the dashed green lines in Figure 1. Notably, there are N_o overlapping cells between the high-elevation-angle cells of adjacent LEO satellites. Define $\mathcal{M}_n$ and $\mathcal{N}_m$ as the set of satellites accessible to the n th cell and the set of cells covered by the m th LEO satellite, respectively.

In the considered system, the network operation control center (NOCC) performs centralized resource management, as depicted in Figure 1, thereby avoiding the synchronization and coordination challenges of distributed computation in dynamic LEO networks. Specifically, each of the M LEO satellites collects the traffic demand of its covered high-elevation-angle cells through signaling channel and transmits this information to the corresponding gateway via the feeder link. The gateways, interconnected through high-speed terrestrial optical fiber networks [23], forward the demand information, satellite positions, and system parameters to the NOCC, which computes the optimal BHTP for the aforementioned LEO satellites. The computed BHTP and user data from corresponding server are then distributed back to each satellite via the gateways for execution. This architecture achieves centralized decision-making and coordination management while eliminating the need for high-speed or low-latency intersatellite links (ISLs). It is assumed that all LEO satellites maintain connectivity to gateways through coordinated handover, which can be ensured via proper gateway deployment and visibility management. The detailed gateway-satellite association mechanism is beyond the scope of this paper.

As mentioned previously, the traffic demand displays uneven spatial and temporal distribution, with each cell having a distinct traffic demand, denoted as D_n . In a beam-hopping system, a single LEO satellite can simultaneously serve up to K cells, employing time-division multiplexing to cover the cells scheduled for service within a beam-hopping cycle. A beam-hopping cycle typically consists of T timeslots, with the illumination patterns of cells in each timeslot collectively forming the beam-hopping pattern. The variable x_{mnt} is defined as an illumination indicator

$$x_{mnt} = \begin{cases} 1, & \text{if satellite } m \text{ illuminates cell } n \text{ at timeslot } t, \\ 0, & \text{otherwise,} \end{cases} \quad (1)$$

To effectively mitigate cofrequency interference among LEO users, this paper assumes that all active beams of a single satellite in a given timeslot provide service to cells using frequency-division multiplexing (FDM), with each beam allocated an equal bandwidth, denoted as B . This assumption aligns with the operational feature of Ku-band satellite systems, such as the downlink beam signals of Starlink.

This paper adopts the widely used free-space path loss channel model [24] to calculate the channel gain between LEO satellite m and user u :

$$h_{mu} = \frac{\sqrt{G_t^L(\theta_{mu})G_r^L(\varphi_{mu})}}{4\pi d_{mu}/\lambda} \quad (2)$$

where $G_t^L(\theta_{mu})$ represents the satellite transmit antenna gain associated with the off-axis angle θ_{mu} , defined as angular deviation of LEO user u from the main lobe direction of LEO satellite m transmit beam; $G_r^L(\varphi_{mu})$ denotes the user receive antenna gain associated with the off-axis angle φ_{mu} , representing angular deviation of LEO satellite m from the main lobe direction of LEO user u receive beam; d_{mu} indicates the distance between satellite m and user u ; and λ signifies the wavelength. And it is assumed that each cell n contains only a single LEO aggregate user u_n^L .

Furthermore, the signal-to-interference-plus-noise ratio (SINR) of LEO user u_n^L at timeslot t can be obtained as

$$\text{SINR}_{mu_n^L t} = \frac{P_{mnt} |h_{mu_n^L t}|^2}{I_{u_n^L} + k_B T_{\text{noise}} B} \quad (3)$$

where P_{mnt} denotes the transmit beam power allocated by LEO satellite m to serve cell n in timeslot t , k_B is the Boltzmann constant, and T_{noise} and B , respectively, represent noise temperature and bandwidth parameters. These three terms can be used to compute noise power at receiver. The interference of LEO user u_n^L , denoted as $I_{u_n^L}$, can be expressed as

$$I_{u_n^L} = P_g |h_{gu_n^L}|^2 \quad (4)$$

where P_g represents the transmission power of GSO satellite g and $h_{gu_n^L}$ denotes the channel coefficient between GSO satellite g and LEO user u_n^L , which can be obtained by substituting the corresponding variables in Equation (2). The interference term in the above expression accounts only for the interference from GSO satellites. This is because intrasatellite cofrequency interference is eliminated by adopting FDM among the active beams of each LEO satellite, while intersatellite cofrequency interference is mitigated through coordinated frequency allocation among adjacent satellites. The use of FDM follows the current practice in Starlink's Ku-band downlink system [25], where active beams are assigned distinct frequency bands to suppress intrasatellite interference under strict antenna sidelobe constraints. Because this paper mainly focuses on the coexistence and interference avoidance between LEO and GSO systems, the analysis considers a frequency reuse factor of one for simplicity. Nevertheless,

the proposed algorithm can be readily extended to higher frequency reuse scenarios, provided that adjacent beams use spatially separated frequencies and the satellite antenna maintains sufficiently low sidelobes. Detailed discussion on these physical-layer aspects is beyond the main scope of this work.

Therefore, the capacity of cell n at timeslot t can be represented as

$$R_{mnt} = B \log_2(1 + \text{SINR}_{mu_n^L t}) \quad (5)$$

Correspondingly, the capacity of cell n over a beam-hopping cycle is given by

$$R_n = \sum_{t=1}^T \sum_{m \in \mathcal{M}_n} x_{mnt} R_{mnt} \quad (6)$$

Moreover, to effectively evaluate the downlink interference experienced by GSO users, this paper adopts the equivalent power flux density (EPFD) metric, as mandated by the ITU. Assuming each cell contains a single GSO user and considering the most severe situation involving colocated LEO and GSO users, the EPFD received by GSO user in the n th cell at timeslot t can be computed as follows:

$$\text{EPFD}_{nt} = 10 \log_{10} \left(\sum_{s \in \mathcal{S}_n} P_{sn,t} \frac{BW_{ref}}{B} \frac{G_t^L(\theta_{su_n^G})}{4\pi d_{su_n^G}^2} \frac{G_r^G(\varphi_{su_n^G})}{G_{r_{max}}^G} \right) \quad (7)$$

where $\mathcal{S}_n$ denotes the set of visible LEO satellites for GSO user in the n th cell, n_s represents the cell served by the visible LEO satellite s , BW_{ref} is the reference bandwidth for EPFD calculation, $d_{su_n^G}$ denotes the distance between LEO satellite s and the GSO user u_n^G , $G_t^L(\theta_{su_n^G})$ represents the LEO satellite transmit antenna gain associated with the off-axis angle $\theta_{su_n^G}$, $G_r^G(\varphi_{su_n^G})$ indicates the GSO user antenna receive gain corresponding to the off-axis angle $\varphi_{su_n^G}$, and $G_{r_{max}}^G$ denotes the maximum receive antenna gain of the GSO user.

It can be observed from the above expression that the interference experienced by the GSO user originates not only from the visible LEO satellites located within the same orbital plane as the serving satellite covering the GSO user, but also from the visible LEO satellites in other orbital planes. Directly applying this expression to constrain the transmit power of active beams would lead to coupling among beam resource allocation schemes of LEO satellites across different orbital planes, which increases the system operational complexity and the coordination overhead among different orbital planes. Considering that the GSO user antenna defined by the ITU for interference evaluation has a relatively narrow main beam, severe interference primarily occurs under colinear or near-colinear conditions. Moreover, because this study focuses on the colocated interference scenario, the set of visible LEO satellites within the analyzed orbital plane causes both mainlobe and sidelobe interference to the GSO user, whereas the visible LEO satellites in other orbital planes contribute only sidelobe interference. Therefore, the aggregate interference received by the GSO user is decomposed into two components: interference from visible LEO satellites within the

analyzed orbital plane and interference from visible LEO satellites in other orbital planes, which effectively decouples the multiplane interference evaluation. To ensure that the total EPFD received by the GSO user always satisfies the ITU-specified limits regardless of the transmit power used by active beams in other orbital planes, the transmit power of those beams is set to its maximum value when computing the EPFD contribution from LEO satellites in other orbital planes.

2.2 | Problem Formulation

Conventional beam-hopping satellite systems primarily focus on metrics such as system capacity, cell traffic demand satisfaction rate, and the vector norm of the difference between cell capacity and demand. However, power is a critical resource for resource-constrained satellite systems. Therefore, this paper adopts RE as the system performance metric, balancing energy efficiency and system capacity normalized by total power, expressed as follows:

$$RE = w \frac{\sum_{n=1}^N \min(R_n, D_n)}{\sum_{t=1}^T \sum_{n=1}^N \sum_{m \in \mathcal{M}_n} (P_{mnt} + x_{mnt} P_c)} + (1-w) \frac{\sum_{n=1}^N \min(R_n, D_n)}{MTP_s + MKTP_c} \quad (8)$$

where $w \in (0, 1)$ denotes the weight factor for the energy efficiency objective, P_c represents the circuit power per beam, and P_s indicates the total beam power per satellite.

To enhance the performance of LEO systems while ensuring interference mitigation to GSO users, this paper formulates a multiobjective optimization problem by optimizing the cell illumination pattern x_{mnt} and the corresponding transmit power P_{mnt} of active beams to maximize system RE. This optimization problem can be formulated as follows:

$$\begin{aligned} \mathcal{P}_0: \quad & \underset{x_{mnt}, P_{mnt}}{\text{maximize}} \quad RE \\ \text{s.t.} \quad & C_1: x_{mnt} P_{b_{min}} \leq P_{mnt} \leq x_{mnt} P_{b_{max}}, \\ & \quad t = 1, \dots, T; \forall m \in \mathcal{M}_n; n = 1, \dots, N \\ & C_2: \sum_{n \in \mathcal{N}_m} P_{mnt} \leq P_s, \quad t = 1, \dots, T; m = 1, \dots, M \\ & C_3: \sum_{n \in \mathcal{N}_m} x_{mnt} \leq K, \quad t = 1, \dots, T; m = 1, \dots, M \\ & C_4: \sum_{m \in \mathcal{M}_n} x_{mnt} \leq 1, \quad t = 1, \dots, T; n = 1, \dots, N \\ & C_5: \text{EPFD}_{nt} \leq \text{EPFD}_{thr}, \quad t = 1, \dots, T; n = 1, \dots, N \end{aligned} \quad (9)$$

C_1 encompasses dual restrictions: First, power allocation is permitted only to scheduled cells; second, the power of a single beam serving a cell is bounded by a maximum limit $P_{b_{max}}$ and a minimum limit $P_{b_{min}}$. C_2 restricts the total power of all activated beams of a single satellite to be below the satellite total beam power P_s . C_3 stipulates that a single satellite can illuminate a maximum of K cells per timeslot. C_4 ensures that each cell receives service at most once per timeslot. C_5 mandates that the

EPFD level received by each GSO user at every time instance remains below the EPFD threshold $EPFD_{thr}$, corresponding to the interference mitigation constraint.

3 | Adjacent Satellites Collaborative Beam-Hopping Interference Avoidance Scheme

It is evident that $\mathcal{P}_0$ constitutes a mixed-integer nonlinear programming problem, considering the nonlinearity of the objective function as well as the need to simultaneously optimize binary variable x_{mnt} and continuous variable P_{mnt} . This represents a typical NP-hard problem that is challenging to solve directly. To address this issue efficiently with reduced complexity, we decompose the original problem into T subproblems, each corresponding to a single timeslot. The subproblem for timeslot t can be formulated as follows:

$$\begin{aligned} \mathcal{P}_1(t): \quad & \underset{x_{mnt}, P_{mnt}}{\text{maximize}} \quad w \frac{\sum_{n=1}^N \min \left(\sum_{m \in \mathcal{M}_n} x_{mnt} R_{mnt} + R_{nt}, D_n \right)}{\sum_{n=1}^N \sum_{m \in \mathcal{M}_n} (P_{mnt} + x_{mnt} P_c) + P_{nt} + P_{c_{nt}}} \\ & + (1-w) \frac{\sum_{n=1}^N \min \left(\sum_{m \in \mathcal{M}_n} x_{mnt} R_{mnt} + R_{nt}, D_n \right)}{MtP_s + MKtP_c} \\ \text{s.t.} \quad & C_1 \sim C_5, \quad \text{applied for timeslot } t. \end{aligned} \quad (10)$$

In the above equations, R_{nt} , P_{nt} , and $P_{c_{nt}}$, respectively, represent the cumulative capacity allocated, the cumulative beam power allocated, and the cumulative circuit power consumed by cell n prior to timeslot t and can be computed individually as follows:

$$\begin{aligned} R_{nt} &= \sum_{\tau=0}^{t-1} \sum_{m \in \mathcal{M}_n} x_{mn\tau} R_{mn\tau} \\ P_{nt} &= \sum_{\tau=0}^{t-1} \sum_{m \in \mathcal{M}_n} P_{mn\tau} \\ P_{c_{nt}} &= \sum_{\tau=0}^{t-1} \sum_{m \in \mathcal{M}_n} x_{mn\tau} P_c \end{aligned} \quad (11)$$

where $R_{mn0} = 0, P_{mn0} = 0, x_{mn0} = 0, \forall m \in \mathcal{M}_n, n = 1, \dots, N$.

Compared with the original problem $\mathcal{P}_0$, the subproblem $\mathcal{P}_1(t)$ effectively reduces the solution space, yet it still involves both discrete variable x_{mnt} and continuous variable P_{mnt} . A common approach to solving such mixed multivariable optimization problems is alternating optimization, which optimizes one variable while fixing the other. This paper adopts this strategy to solve subproblem $\mathcal{P}_1(t)$. Specifically, we first employ a greedy algorithm prioritizing residual demand under adaptive demand satisfaction ratio constraints to obtain the solution for the timeslot allocation variable x_{mnt}^* . Subsequently, with the timeslot allocation fixed, we employ a quadratic transformation to convert the concave-convex fractional programming problem into a convex optimization problem and utilize the CVX toolbox to obtain the solution of power allocation variable P_{mnt}^* . After obtaining the user scheduling and power allocation for a single timeslot using these steps, the variables R_{nt} , P_{nt} , and $P_{c_{nt}}$ are updated, and the process is repeated to solve the subproblem for

the next timeslot. This iterative approach across timeslots ultimately yields a locally optimal solution of the original problem $\mathcal{P}_0$. And the details of the aforementioned adjacent satellites collaborative beam-hopping interference avoidance technique are presented below.

3.1 | Adaptive Demand Satisfaction Ratio User Grouping and Residual Demand Priority Adjacent Satellites Collaborative Greedy User Scheduling

Regarding the composition of the objective function, RE is a weight sum of energy efficiency and system capacity normalized by total power, where the weight factor w directly determines the inclination of RE toward the energy efficiency objective, with a larger w value shifting the optimization of RE closer to that of energy efficiency. To enable the user scheduling algorithm to adapt to objective functions with varying inclinations, this paper proposes an adaptive demand satisfaction ratio grouping strategy. Specifically, we first select from all cells those with a cumulative capacity R_{nt} allocated prior to the current timeslot t that is below D_n to form a set of cells to be served, denoted as $\mathcal{N}_{re}$, to avoid beam power wastage. Then, from $\mathcal{N}_{re}$, we select cells where R_{nt} is below the capacity corresponding to the adaptive demand satisfaction ratio αD_n to form a subset $\mathcal{N}_{re1}$. The remaining cells in $\mathcal{N}_{re}$ constitute the subset $\mathcal{N}_{re2}$.

After obtaining the two subsets of cells to be served, we first sort the cells in $\mathcal{N}_{re1}$ in descending order based on the residual demand $\alpha D_n - R_{nt}$ corresponding to the adaptive demand satisfaction ratio. Next, we sort the cells in $\mathcal{N}_{re2}$ in descending order based on the residual demand $D_n - R_{nt}$. Finally, the two sorted subsets are merged, ensuring that cells in $\mathcal{N}_{re1}$, after sorting, retain higher service priority, forming a candidate set of cells with scheduling priority, denoted as $\mathcal{N}'_{re}$.

To obtain the user scheduling solution x_{mnt}^* for the current timeslot t , we sequentially attempt to schedule cells from the candidate set $\mathcal{N}'_{re}$ in order of their priority according to a greedy strategy until all satellites exhaust their current scheduling resources or all cells in $\mathcal{N}'_{re}$ have been considered. For each cell, the adjacent satellites collaborative serving approach proposed in this paper enables multiple accessible satellites, and a greedy access strategy is thus employed to select the optimal satellite that satisfies interference mitigation constraints while maximizing RE. Specifically, for the cell n under consideration, the process begins by selecting from the set of candidate satellites $\mathcal{M}_n$ those satellites outside the main lobe of GSO user receive antennas to form a low-interference candidate set $\mathcal{M}'_n$, significantly reducing interference to GSO users. Subsequently, within $\mathcal{M}'_n$, satellites with residual scheduling resources in the current timeslot are attempted for access in ascending order of distance from the LEO user until a successful connection is established or all satellites have been tried, thereby enhancing energy efficiency and increasing system capacity to maximize the objective function. Additionally, for cells in overlapping regions with two equidistant nearest satellites, priority is given to the satellite with greater remaining scheduling resources in the current timeslot to ensure service for more users.

3.2 | Quadratic Transformation-Based Convex Optimization Power Allocation

Following the determination of user scheduling x_{mnt}^* for the current timeslot, optimization of power allocation for user beams becomes necessary. When the energy efficiency weight factor w is excessively large, the RE objective of subproblem $\mathcal{P}_1(t)$ may lead to power allocation results favoring extreme minimum power allocations, neglecting system traffic demands, which is undesirable. To address this, this paper incorporates the aforementioned adaptive demand satisfaction ratio corresponding to traffic demand αD_n as a constraint penalty term, with penalty factor ρ into the RE objective function, formulating a new subproblem $\mathcal{P}_2(t)$. The subproblem at timeslot t can be expressed as follows:

$$\begin{aligned} \mathcal{P}_2(t): \quad & \underset{P_{mnt}}{\text{maximize}} \quad w \frac{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt}, D_n)}{\sum_{n=1}^N (P_{mnt} + x_{mnt}^* P_c) + P_{nt} + P_{c_{nt}}} \\ & + (1-w) \frac{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt}, D_n)}{MtP_s + MKtP_c} \\ & + \rho \frac{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt} - \alpha D_n, 0)}{MtP_s + MKtP_c} \\ \text{s.t.} \quad & C'_1: x_{mnt}^* P_{b_{\min}} \leq P_{mnt} \leq x_{mnt}^* P_{b_{\max}}, \quad n=1, \dots, N \\ & C'_2: \sum_{n \in \mathcal{N}_m} P_{mnt} \leq P_s, \quad m=1, \dots, M \\ & C'_3: \text{EPFD}_{nt} \leq \text{EPFD}_{thr}, \quad n=1, \dots, N \end{aligned} \quad (12)$$

Noting that the energy efficiency term in the objective function of subproblem $\mathcal{P}_2(t)$ constitutes a typical fractional programming problem with respect to the power variable P_{mnt} , which is challenging to solve directly, existing algorithms for addressing fractional programming problems include the Charnes-Cooper transform, Dinkelbach's transform, and quadratic transform [26]. Among these, the latter two methods do not require the introduction of additional constraints, while the quadratic transform ensures that the optimal value of the fractional programming objective remains invariant before and after transformation. This property is critical for the multiobjective subproblem $\mathcal{P}_2(t)$, as it guarantees the consistency of the overall objective function's optimal value across the transformation; thus, this paper adopts this approach to derive the power allocation scheme for the current timeslot. Based on the implementation of the quadratic transform, we introduce an auxiliary variable y to reformulate the energy efficiency term of the objective function, enabling the new subproblem $\mathcal{P}'_2(t)$ to be expressed as follows:

$$\begin{aligned} \mathcal{P}'_2(t): \quad & \underset{P_{mnt}, y}{\text{maximize}} \quad 2wy \sqrt{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt}, D_n)} \\ & - wy^2 \left(\sum_{n=1}^N (P_{mnt} + x_{mnt}^* P_c) + P_{nt} + P_{c_{nt}} \right) \\ & + (1-w) \frac{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt}, D_n)}{MtP_s + MKtP_c} \\ & + \rho \frac{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt} - \alpha D_n, 0)}{MtP_s + MKtP_c} \\ \text{s.t.} \quad & C'_1 \sim C'_3. \end{aligned} \quad (13)$$

We address subproblem $\mathcal{P}'_2(t)$ using an iterative alternating optimization approach; specifically, given P_{mnt} , the auxiliary variable y is updated according to the following equation:

$$y^* = \frac{\sqrt{\sum_{n=1}^N \min(x_{mnt}^* R_{mnt} + R_{nt}, D_n)}}{\sum_{n=1}^N (P_{mnt} + x_{mnt}^* P_c) + P_{nt} + P_{c_{nt}}} \quad (14)$$

The update mechanism for the auxiliary variable y constitutes a critical step in ensuring the equality of the optimal value between the objective functions of subproblems $\mathcal{P}_2(t)$ and $\mathcal{P}'_2(t)$. When y is fixed, the objective function of $\mathcal{P}'_2(t)$ becomes a concave function with respect to the power variable P_{mnt} , attributable to the concavity of the square root operation, the minimum operation, and the logarithmic terms within R_{mnt} , which collectively preserve concavity under composition.

Furthermore, the single-beam power constraint and the per-satellite power constraint are both convex constraints. Notably, given that the current user scheduling x_{mnt}^* is determined, the interference mitigation requirements impose convex constraints on beam power. Consequently, with y fixed, subproblem $\mathcal{P}'_2(t)$ corresponds to the problem of maximizing a concave objective function subject to convex constraints, a standard convex optimization problem that can be efficiently solved using the CVX toolbox. The iterative alternating optimization process, alternating between updating the auxiliary variable y and the power variable P_{mnt} , is continued until the objective function converges or a preset maximum number of iterations is reached, thereby completing the solution of subproblem $\mathcal{P}'_2(t)$. As previously noted, owing to the equality of the optimal values of the objective functions of $\mathcal{P}_2(t)$ and $\mathcal{P}'_2(t)$, and their attainment at the same P_{mnt}^* points, this approach effectively yields the optimal power allocation solution for subproblem $\mathcal{P}'_2(t)$.

Integrating the methods outlined in the preceding two subsections, the local optimal solution of the original problem $\mathcal{P}_0$ can be obtained by iteratively solving for user scheduling x_{mnt} and power allocation P_{mnt} timeslot by timeslot, while updating the variables R_{nt} , P_{nt} , and $P_{c_{nt}}$. The detailed implementation process is presented in Algorithm 1.

The computational complexity of the proposed adjacent satellites collaborative beam-hopping interference avoidance scheme is analyzed as follows. For each timeslot, the user grouping and priority sorting (Lines 3–7) is dominated by the sorting operation, yielding a complexity of $O(N \log N)$, where N is the maximum number of cells with unsatisfied traffic demand. The greedy user scheduling (Lines 8–16) iterates through the sorted cell set $\mathcal{N}'_{re}$ with at most N cells, and because each cell has at most three candidate satellites, the complexity is $O(N)$ in the worst case. The power allocation via quadratic transformation (Lines 17–21) requires iteratively solving a convex subproblem by CVX, whose complexity is approximately $O(I_1(MK)^3)$, where I_1 denotes the maximum iteration count of quadratic transform, M is the number of satellites, and K is the maximum number of active beams per satellite. Therefore, the overall computational complexity of the proposed algorithm across T timeslots can be expressed as $O(T(N \log N + N + I_1(MK)^3))$, which remains computationally tractable because I_1 is generally small.

To further improve scalability for large-scale LEO constellations, the power allocation subproblem (Line 19), which dominates the overall computational complexity, can alternatively be solved using the alternating Lagrangian-dual iterative method, with a maximum iteration count of I_2 . When Lagrangian multipliers are fixed, the constructed Lagrangian function does not have a closed-form solution for the optimal value when maximizing the function because of its complexity. Therefore, we apply the golden section search to obtain a near-optimal value, with a maximum iteration count of I_3 . Additionally, considering the coupling of multiple power variables in the first term of the original objective function of the subproblem, we use the first-order Taylor expansion to approximate the function, enabling decoupling of variables and reducing the computational complexity to linear. Due to the maximum iteration count set for the aforementioned alternating Lagrangian-dual iterative algorithm, the obtained solution may not strictly satisfy the constraints, which can be addressed by applying a heuristic projection with a maximum iteration count of I_4 . The final complexity of the alternating Lagrangian-dual method to solve the subproblem can thus be expressed as $O((I_2I_3 + I_4)MK)$, which is significantly lower than the CVX-based approach with complexity $O((MK)^3)$.

The overall complexity of the proposed algorithm, considering all steps including user grouping, sorting, scheduling, and power allocation for each timeslot, can be expressed as $O(T(N \log N + N + I_1(MK)^3))$ for the CVX-based implementation, and $O(T(N \log N + N + I_1((I_2I_3 + I_4)MK)))$ for the Lagrangian-dual implementation. The latter exhibits linearithmic complexity with respect to the number of cells and linear complexity with respect to the number of satellites. This makes the method highly scalable and computationally efficient for large-scale LEO constellations, while still achieving near-optimal performance close to the CVX-based implementation, as demonstrated in the simulation results in the next section.

It should be noted that, for analytical tractability, the transmit power of each active beam in this work is assumed to be continuously adjustable within its allowable range. In practical satellite communication systems, however, power control is typically implemented in discrete steps due to hardware and amplifier design constraints. Such discretization may cause a minor deviation when mapping the theoretical continuous results to actual implementations. Nevertheless, this assumption does not affect the generality of the proposed optimization framework, and a discrete power adjustment model will be investigated in future work to further enhance engineering applicability.

4 | Simulation Results

In simulation, we study the coexistence scenario of partial LEO satellites in a single orbital plane of Walker constellation and GSO satellites under downlink operation, as depicted in Figure 2, to assess the effectiveness of the proposed scheme. Table 1 lists the simulation parameters for the two satellite systems, including orbital parameters and communication parameters. The orbital parameters are derived from the current common constellation orbital parameters [27], while the communication parameters are based on relevant communication

Algorithm 1. Adjacent satellites collaborative beam-hopping interference avoidance scheme.

Input: Number of satellites M ; total cells N ; timeslots T ; beams per satellite K ; traffic demand D_n for cell n ; system parameters; adaptive factor α ; satellite set $\mathcal{M}_n$ for cell n

Output: Scheduling matrix x_{mnt}^* and power allocation matrix P_{mnt}^*

```

1: Initialize  $R_{nt} \leftarrow 0$ ,  $P_{nt} \leftarrow 0$ ,  $P_{c_{nt}} \leftarrow 0$ 
2: for  $t = 1$  to  $T$ 
    // Step 1: User Grouping and Priority Sorting
3:   Select cells with  $R_{nt} < D_n$  to form set  $\mathcal{N}_{re}$ 
4:   From  $\mathcal{N}_{re}$ , select cells with  $R_{nt} < \alpha D_n$  to form  $\mathcal{N}_{re1}$ 
5:   Compute  $\mathcal{N}_{re2} = \mathcal{N}_{re} \setminus \mathcal{N}_{re1}$ 
6:   Sort  $\mathcal{N}_{re1}$  and  $\mathcal{N}_{re2}$  by descending  $\alpha D_n - R_{nt}$  and  $D_n - R_{nt}$ , respectively
7:   Concatenate sorted  $\mathcal{N}_{re1}$  and  $\mathcal{N}_{re2}$  to form  $\mathcal{N}'_{re}$ 
    // Step 2: Greedy User Scheduling
8:   for each  $n \in \mathcal{N}'_{re}$ 
9:     From  $\mathcal{M}_n$ , exclude satellites in GSO main lobe to form  $\mathcal{M}'_n$ 
10:    Sort  $\mathcal{M}'_n$  by ascending distance to cell  $n$ 
11:    for each  $m \in \mathcal{M}'_n$ 
12:      if satellite  $m$  has residual scheduling resources
13:        Set  $x_{mnt}^* \leftarrow 1$  and break
14:      end if
15:    end for
16:  end for
    // Step 3: Power Allocation via Quadratic Transform
17:  Initialize auxiliary variable  $y$  via (14)
18:  repeat
19:    Fix  $y$ , solve (13) using CVX to obtain exact optimal  $P_{mnt}$ , or using alternating Lagrangian-dual iterative method to obtain near-optimal  $P_{mnt}$ 
20:    Fix  $P_{mnt}$ , update  $y$  using (14)
21:  until convergence or reaching max iterations  $I_1$ 
    // Step 4: Update
22:  Update  $R_{nt}$ ,  $P_{nt}$ ,  $P_{c_{nt}}$  using (11)
23: end for
24: return  $x_{mnt}^*$ ,  $P_{mnt}^*$ 

```

standards and the existing satellite communication parameters of current constellations [28]. Specifically, the beam-hopping frame consists of 64 timeslots of 10ms each, yielding a total

TABLE 1 | Simulation parameters.

Parameters	Value
Scenario parameters	
Earth radius	6378.14 km
Minimum downlink frequency	10.7 GHz
Maximum downlink frequency	12.7 GHz
Beam bandwidth	250 MHz
Noise temperature	290 K
GSO system parameters	
Orbit altitude	35,786 km
Inclination	0°
Total number of satellites	180
Total number of planes	1
Phasing factor	0
Minimum elevation angle	15°
Satellite EIRP density	46 dBW/MHz
Peak gain of GSO satellite antenna	52 dBi
GSO user receiving antenna diameter	60 cm
LEO system parameters	
Orbit altitude	1150 km
Inclination	88.4°
Total number of satellites	720
Total number of planes	20
Phasing factor	10
Number of satellites in simulation	15
Number of high-elevation-angle cells	91
Number of overlapping cells	13
Total number of cells in simulation	1183
Peak gain of LEO satellite antenna	28 dBi
LEO user receiving antenna diameter	40 cm
Maximum power of LEO beam	8 W
Minimum power of LEO beam	1 W
Total power of LEO satellite	60 W
The maximum beams per satellite	8
Timeslot duration	10 ms
Number of timeslots	64
EPFD parameter	
EPFD limit	-175.4 dB(W/m ² /40 kHz)

cycle of 640 ms. This setting, derived from the DVB-S2X beam-hopping protocol [29] and widely adopted in existing studies [30–32], ensures both modeling consistency and practical feasibility. It provides sufficient temporal granularity for adaptive resource allocation within the coordinated orbital plane while remaining compatible with the centralized NOCC-based scheduling framework. Given that user traffic and channel variations evolve on much slower timescales, the latency introduced by centralized computation and signaling is negligible.

In addition, to effectively evaluate the performance of the interference avoidance algorithm, the EPFD limit is listed in the table of simulation parameters. Although the EPFD limits specified in Article 22 of the ITU Radio Regulations are not defined as a single limit but rather as a set of time proportion dependent constraints, in practice, related research papers and even technical documents from satellite operators generally adopt a single EPFD limit when designing interference avoidance strategies. For example, Hills et al. [6] used the relatively relaxed limit corresponding to 100 % of the time to specify the transmit antenna sidelobe levels that must be satisfied by the OneWeb and Starlink satellite systems, respectively. Telesat, in its technical report [33], adopted the limit for 0 % of the time to calculate the minimum discrimination angle for the Polar Orbits in the 17.8- to 18.6-GHz band. Directly using a single limit simplifies the design of interference avoidance strategies. Moreover, by selecting the EPFD limit corresponding to 0 % of the time as the constraint for interference mitigation design, the ITU requirements can still be strictly satisfied even when considering the time statistical characteristics of EPFD. For this reason, this paper adopts the strictest EPFD limit for 0 % of the time to fully ensure the interference avoidance performance of the proposed algorithm. This consideration is consistent with that of Jalali et al. [34], who derived the transmit power parameters and tilt angles for their proposed joint power and tilt control technique based on the same EPFD limit.

It should also be noted that adopting the strictest EPFD constraint -175.4 dB(W/m²/40 kHz) inevitably leads to a conservative design that may impose overprotection on GSO users and restrict the transmit power of the LEO system. In fact, a margin of approximately 15 dB exists between the -175.4- and -160-dB(W/m²/40 kHz) limits defined by ITU Article 22, which represents a potential performance space that remains unutilized. This trade-off was intentionally adopted in this study to guarantee complete regulatory compliance, while an adaptive EPFD margin adjustment strategy will be investigated in future work to achieve a more balanced compromise between interference protection and transmission efficiency.

We set the number of LEO satellites in simulation to $M = 15$, considering the GSO satellite coverage corresponding to the minimum elevation angle for communication and the orbital parameters of LEO constellation. As previously noted, a single satellite typically serves $N_h = 91$ high-elevation-angle cells, whereas the proposed method assumes that each satellite can serve the $N_h = 91$ high-elevation-angle cells of both itself and adjacent satellites. Additionally, due to the overlap of $N_o = 13$ cells between the high-elevation-angle cells of adjacent satellites, the 15 satellites exhibit the following service distribution: the two end satellites each serve $2N_h - N_o = 169$ cells,

while the remaining satellites each serve $3N_h - 2N_o = 247$ cells. Furthermore, the overall count of cells served by the 15 satellites can be calculated as $MN_h - (M - 1)N_o = 1183$ cells.

Each satellite is capable of forming a maximum of eight spot beams per timeslot. Without loss of generality, LEO users are assumed to be located at the center of their respective cells and are oriented toward the LEO satellite currently providing service, while the LEO satellite directs its spot beams toward the center of the currently served cell to simplify the problem analysis. As previously noted, this paper considers a colocated interference scenario [35]; thus, GSO users within each cell are also positioned at the corresponding cell center, with their receive antennas oriented toward the GSO satellite, ensuring that the proposed technique effectively mitigates interference under all conditions.

The current spectrum coexistence scenario encompasses three distinct antenna radiation patterns. According to 3GPP TR 38.811 [36], we adopt the following model to calculate the LEO satellite transmit antenna beam pattern:

$$G_t^L(\theta) = \begin{cases} G_{\max}, & \theta = 0^\circ \\ 4G_{\max} \left| \frac{J_1(2\pi a \sin \theta / \lambda)}{2\pi a \sin \theta / \lambda} \right|^2, & 0^\circ < |\theta| \leq 90^\circ \end{cases} \quad (15)$$

where λ is the wavelength, a denotes the radius of the antenna circular aperture, $J_1(x)$ represents the first-kind first-order Bessel function with parameter x , θ denotes the off-axis angle of the antenna beam, and $G_{\max}$ represents the maximum antenna gain, calculated as $G_{\max} = 20 \log(2\pi a / \lambda)$. The GSO satellite transmit antenna beam pattern is referenced from ITU-R S.672-4 [37] and can be computed from the following model:

$$G_t^G(\theta) = \begin{cases} G_{\max} - 3 \left(\frac{\theta}{\theta_b} \right)^2, & 0 \leq \theta \leq 2.58\theta_b \\ G_{\max} - 20, & 2.58\theta_b < \theta \leq 6.32\theta_b \\ G_{\max} - 20 - 25 \log \left(\frac{\theta}{6.32\theta_b} \right), & 6.32\theta_b < \theta \leq 6.32\theta_b 10^{0.04(G_{\max} - 20)} \\ 0, & 6.32\theta_b 10^{0.04(G_{\max} - 20)} < \theta \leq 90^\circ \end{cases} \quad (16)$$

where G_m represents the maximum radiation gain, calculated as $G_m = 20 \log(D / \lambda) + 7.7$, with D representing the antenna diameter, θ_b denotes half of the 3-dB beamwidth. The receive antennas of both LEO and GSO users employ the ITU-R S.1428-1 [38] antenna model, which can be expressed as follows:

$$G_r(\varphi) = \begin{cases} G_{\max} - 2.5 \times 10^{-3} \left(\frac{\varphi D}{\lambda} \right)^2, & 0 \leq \varphi \leq \varphi_m \\ G_1, & \varphi_m < \varphi \leq \frac{95\lambda}{D} \\ 29 - 25 \log \varphi, & \frac{95\lambda}{D} < \varphi \leq 33.1^\circ \\ -9, & 33.1^\circ < \varphi \leq 80^\circ \\ -5, & 80^\circ < \varphi \leq 90^\circ \end{cases} \quad (17)$$

where

$$\begin{aligned} G_{\max} &= 20 \log \left(\frac{D}{\lambda} \right) + 7.7 \\ G_1 &= 29 - 25 \log \left(\frac{95\lambda}{D} \right) \\ \varphi_m &= \left(\frac{20\lambda}{D} \right) \sqrt{G_{\max} - G_1} \end{aligned} \quad (18)$$

Figure 3 presents the antenna gain patterns for the aforementioned LEO satellites, GSO satellites, and their corresponding users.

The user traffic demand is correlated with the geographical distribution of population [39], typically concentrated in mid-to-low latitude regions. Based on this, we generate the traffic demand D_n for individual cell, as illustrated in the first subfigure of Figure 4. Notably, the cell indices commence from the northernmost boundary of the GSO satellite coverage area, extending to the southernmost boundary, with the low-latitude regions in the intermediate area corresponding to higher traffic demand. This traffic distribution characteristic is also

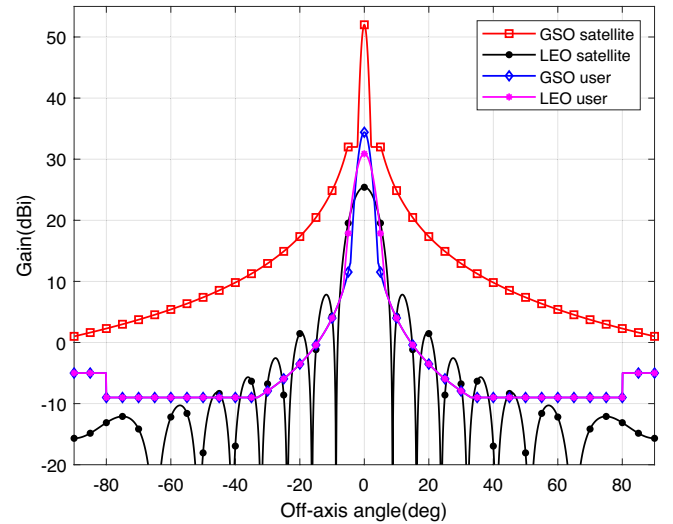


FIGURE 3 | Antenna gain patterns.

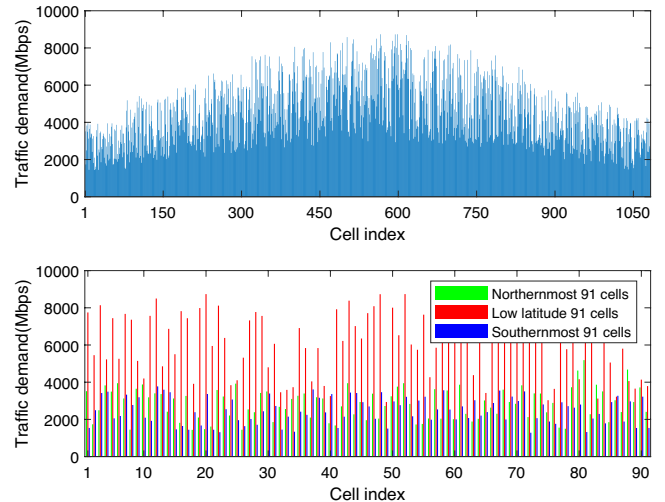


FIGURE 4 | The traffic demand of different cells.

reflected in the second subfigure of Figure 4, which compares the traffic demand of the 91 high-elevation-angle cells covered by the northernmost satellite and the southernmost satellite with those of the 91 high-elevation-angle cells covered by satellite at low latitude.

In this paper, we select four representative beam-hopping schemes, along with the proposed method, for performance comparison. The four comparative schemes are outlined as follows:

1. SINR maximization beam hopping (SINR Max-BH): As reported in the literature [40], the SINR Max-BH method determines beam slot allocation through pre-assigned timeslots and a criterion prioritizing the fulfillment of lower demand proportions. Subsequently, the scheduled cells select the serving satellite using the maximum elevation angle access method to ensure SINR performance. Finally, system capacity is further enhanced through optimized power allocation.
2. Greedy-based beam hopping (G-BH): The G-BH approach [41] means each satellite selects top 8 demand-unsatisfied cells with the highest traffic demand within its coverage area per timeslot, while distributing satellite power uniformly across the activated beams.
3. Joint iterative multisatellite beam hopping (JIMS-BH): The JIMS-BH method [42] comprises two steps: First, it addresses the cell and satellite assignment problem to ensure balanced traffic demand loads across satellites, while mitigating interference to GSO users by constraining the interference link angles; second, it iteratively selects the timeslot exhibiting the least overall interference to serve the cell whose traffic satisfaction rate is lowest, with satellite power uniformly allocated to the activated beams.
4. Genetic algorithm beam hopping (GA-BH): This method employs a genetic algorithm, as proposed in the literature [43], that considers cell fairness, optimizing and iteratively updating cell selection and beam power timeslot by timeslot. Additionally, the simulation incorporates constraints on interference link angles to avoid interference with GSO users.

As mentioned earlier, the Lagrangian-dual implementation of the proposed algorithm uses heuristic projection to strictly satisfy all constraints, including the EPFD limit, ensuring the interference avoidance performance of the method. Therefore, the following simulations will solely include a comparative analysis of the communication performance of this method.

To effectively assess the interference mitigation capability of various beam-hopping schemes, we present the complementary cumulative distribution function (CCDF) curves of GSO user received EPFD levels over the beam-hopping cycle, as depicted in Figure 5. It is evident that, aside from the SINR Max-BH and G-BH schemes, which do not account for downlink interference to GSO users, all other beam-hopping schemes employing interference avoidance strategies significantly reduced the interference to GSO users, and the CCDF curves of the received EPFD levels shifted left accordingly. Among them, only the proposed

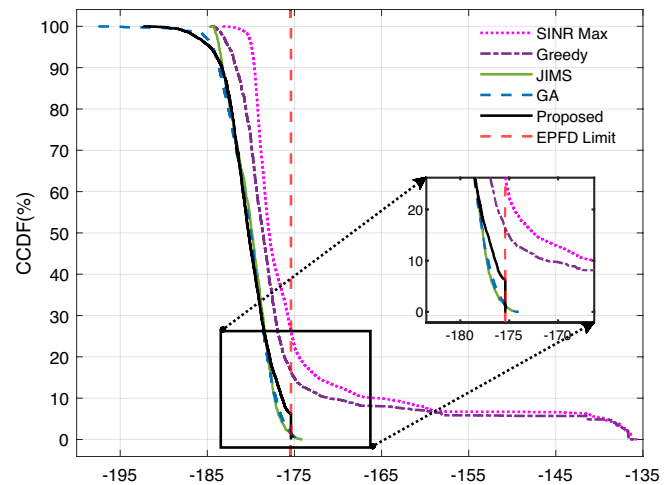


FIGURE 5 | The CCDF of GSO user received EPFD levels for different beam-hopping schemes across cells.

method fully satisfies the strictest EPFD limit corresponding to 0% of the time as specified by the ITU. This is because the proposed algorithm jointly optimizes the interference link angle and transmit power parameters by considering both the constraints associated with the aforementioned limit and the aggregate sidelobe interference from the set of visible LEO satellites in other orbital planes, thereby effectively ensuring interference avoidance performance for all GSO users.

For a detailed analysis of the spatial distribution of interference levels, we select the SINR Max-BH scheme and the proposed beam-hopping scheme as representatives, presenting the average GSO user received EPFD levels over the beam-hopping cycle for mid-to-low latitude regions in the northern and southern hemispheres in Figure 6a,b. As observed in Figure 6a, the downlink interference from LEO satellites to GSO users exhibits a strong correlation with the latitude of the cells where GSO users are located, with GSO users in low-latitude cells experiencing more severe interference. Notably, even within low-latitude cells, GSO users near the equator receive greater interference, with average EPFD levels exceeding $-140\text{ dB(W/m}^2/40\text{ kHz)}$, more than 30 dB above the ITU limit, due to their colinear or near-colinear alignment with LEO and GSO satellites. Additionally, cells where interference exceeds the ITU limit display a distinct banded distribution near the equator, with these interference bands curving concavely toward the equator in the northern and southern low-latitude regions, a phenomenon also attributable to colinear or near-colinear spatial relationships. In contrast, the proposed method enables more flexible interference mitigation: It increases the interference link angle through adjacent satellite cooperation in colinear regions, while adjusting transmit power in noncolinear regions. Together, these strategies ensure that the GSO user received EPFD levels remain below the ITU limit across all regions, effectively meeting the interference mitigation requirements, as demonstrated in Figure 6b.

Beyond the aforementioned interference mitigation performance, we assess the performance of the LEO satellite system for various beam-hopping methods using multiple metrics. It should be noted that other beam-hopping schemes, as detailed in the respective literature, are optimized for single

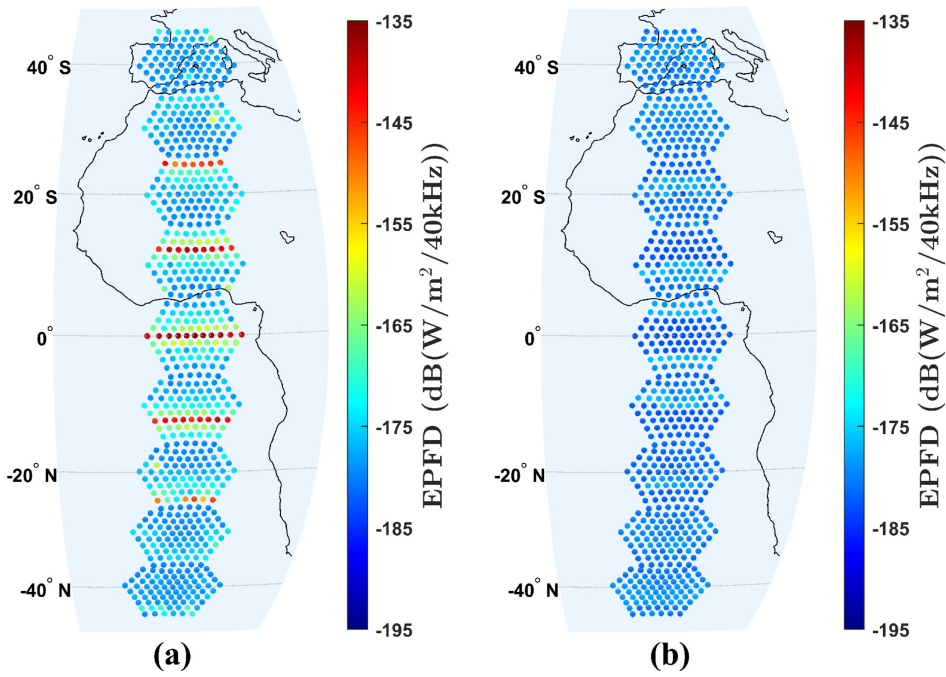


FIGURE 6 | Average received EPFD levels of different GSO users over the beam-hopping cycle: (a) the SINR Max-BH and (b) the proposed beam hopping.

communication performance metrics, such as the minimum demand satisfaction rate, the sum of system capacities across all cells, or the sum of demand satisfaction rate across all cells, exhibiting uniform communication performance across different energy efficiency weight factors w . These schemes differ only in RE performance, which is influenced by the energy efficiency weight factor w . In contrast, the proposed beam-hopping scheme incorporates an adaptive demand satisfaction ratio operation, enabling it to adapt to the preference degree of the RE objective function toward energy efficiency. Consequently, the proposed scheme demonstrates varying communication performance across different energy efficiency weight factors w , effectively balancing energy efficiency metrics and system capacity metrics. Without loss of generality, subsequent simulations adopt $w = 0.7$ and $w = 0.3$ to represent RE objective functions favoring energy efficiency and system capacity, respectively.

Figure 7 illustrates the cumulative distribution function (CDF) of the demand satisfaction rate across all cells under different beam-hopping schemes, where the demand satisfaction rate for cell n is defined as $\gamma_n = R_n/D_n$. It is apparent that, regardless of the energy efficiency weight factor w , the proposed beam-hopping scheme achieves superior performance in both implementations, with the minimum cell demand satisfaction rate exceeding 90% and higher than that of other schemes. The CVX-based implementation attains the lowest overservice rate and proportion of overserved cells, reducing capacity waste and maintaining optimal performance, thus representing the upper performance bound of the proposed scheme. In contrast, the Lagrangian-dual implementation performs between the JIMS and GA beam-hopping schemes and shows slight power and capacity inefficiencies. This is mainly because three complexity-reduction strategies—limiting the number of iterations, employing numerical solution methods, and applying

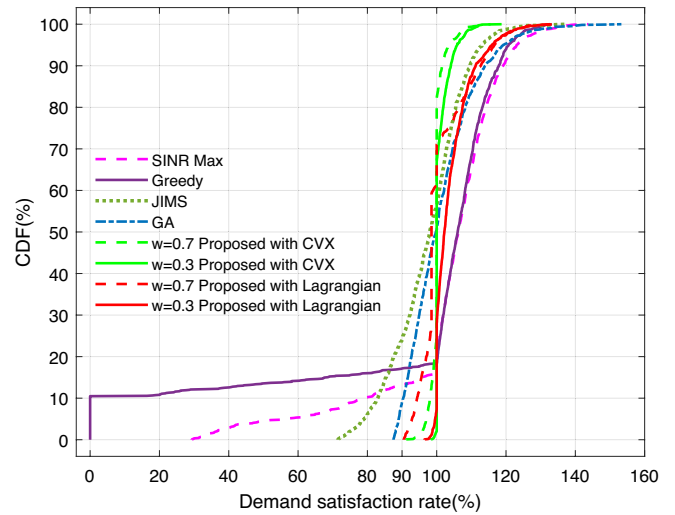


FIGURE 7 | The CDF of cell demand satisfaction rate for different beam-hopping schemes.

convex approximation to decouple multiple power variables—effectively lower computational complexity but also reduce the precision in solving the power variables P_{mnt} during each quadratic transform iteration. Moreover, both implementations exhibit a consistent trend with respect to the energy efficiency weight factor w : a larger w improves energy efficiency, whereas a smaller w emphasizes system capacity. As previously noted, the CVX-based implementation achieves higher precision in solving the power variables, thereby showing better adaptability to variations in w .

Additionally, we compare the performance for different beam-hopping schemes with respect to system RE and the total capacity across all cells, as illustrated in Figure 8. On one hand,

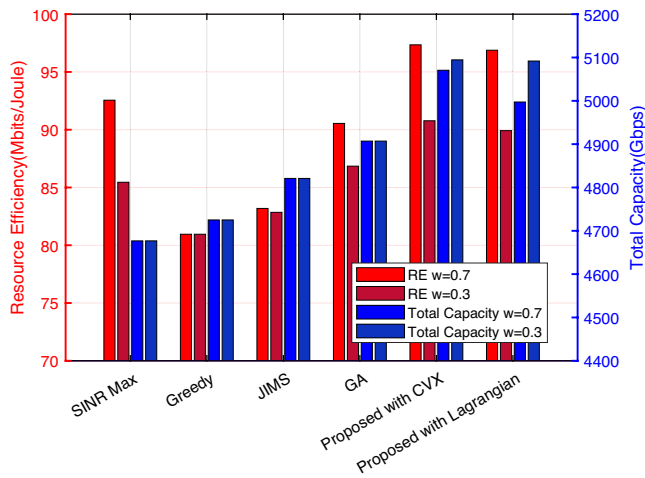


FIGURE 8 | The comparison of RE and total capacity of all cells for different beam-hopping schemes.

for all beam-hopping schemes, a larger energy efficiency weight factor w corresponds to higher RE, a characteristic determined by the composition of the objective function. As expressed in Equation (8), the objective function comprises a weight sum of the energy efficiency term and the normalized system capacity term with respect to total power, where the numerators of both fractional terms are the total capacity across all cells, while the denominator of the former, representing the actual total power allocated by the system, is less than or equal to the denominator of the latter, corresponding to the system's maximum available total power limit. Consequently, RE exhibits a positive correlation with the energy efficiency weight factor w . On the other hand, as previously noted, the proposed beam-hopping scheme, in both its CVX-based and Lagrangian-dual implementations, demonstrates varying communication performance under different energy efficiency weight factors w , achieving a greater total capacity across all cells when the weight factor is smaller, whereas the other beam-hopping schemes maintain a uniform total capacity across all cells regardless of the weight factor. Furthermore, under varying energy efficiency weight factors, the proposed beam-hopping scheme based on the CVX implementation exhibits optimal performance in both RE and total capacity across all cells, while the Lagrangian-dual implementation achieves near-optimal performance in both metrics. Specifically, the RE of the proposed scheme based on the CVX implementation improves by 5.2%, 20.2%, 17.0%, and 7.5% and by 4.7%, 19.7%, 16.5%, and 7.0%, respectively, compared with the SINR Max, Greedy, JIMS, and GA beam-hopping schemes at $w = 0.7$, and by 6.2%, 12.1%, 9.6%, and 4.5% and by 5.2%, 11.1%, 8.5%, and 3.5%, respectively, at $w = 0.3$. Moreover, the total capacity across all cells for the proposed scheme based on the CVX implementation and that based on the Lagrangian-dual implementation improves by 8.4%, 7.3%, 5.2%, and 3.3% and by 6.9%, 5.8%, 3.7%, and 1.8%, respectively, compared with the SINR Max, Greedy, JIMS, and GA beam-hopping schemes at $w = 0.7$ and by 8.9%, 7.8%, 5.7%, and 3.8% and by 8.9%, 7.8%, 5.6%, and 3.8%, respectively, at $w = 0.3$. As previously discussed, the performance difference between the two implementations of the proposed beam-hopping scheme arises from the lower solution accuracy of the power variables in the

Lagrangian-dual implementation, which leads to power and capacity wastage. These simulation findings further confirm the efficacy of the proposed beam-hopping method. Overall, the proposed scheme achieves superior total capacity and energy efficiency performance compared with the four benchmark algorithms under identical conditions, demonstrating a clear advantage in adaptive interference-aware scheduling for LEO–GSO coexistence.

The residual traffic demand of cells provides a direct reflection of the demand satisfaction status across different beam-hopping schemes, where the residual traffic demand for cell n is defined as $\max(D_n - R_n, 0)$. It should be noted that the first two sets of simulation results and analyses have demonstrated that the proposed algorithm with the Lagrangian-dual implementation can achieve near-optimal performance comparable to the CVX-based implementation. Therefore, considering the paper length and the size of the simulation figures, when analyzing the remaining traffic demand of cells under different beam-hopping schemes, only the simulation results of the proposed beam-hopping scheme based on the CVX implementation are compared with those of other beam-hopping schemes. To facilitate the visualization of residual traffic demand across cells, we present the residual traffic demand of 13 cells per column in a column-priority order, as depicted in Figure 9. The SINR Max-BH scheme, employing a maximum elevation angle access rule, effectively serves low-traffic-demand cells at high latitude but fails to meet the demands of high-traffic-demand cells at low latitude. The G-BH scheme prioritizes serving high-traffic-demand cells, resulting in substantial residual traffic demand for the majority of low-traffic-demand cells at high latitude and a minority of low-traffic-demand cells at low latitude, which are inadequately served. Furthermore, as previously noted, these two beam-hopping schemes do not account for mitigating interference to GSO users, consequently reducing the communication rates of LEO users and resulting in substantial residual traffic demand, particularly in low-latitude cells with higher interference risk. The JIMS-BH scheme allocates satellites and cells in a load balancing manner: low-latitude satellites serve fewer high-traffic-demand cells over the beam-hopping cycle, while mid-to-high-latitude satellites serve a greater number of low-traffic-demand cells. Additionally, interference mitigation measures that constrain interference link angles further increase the number of cells requiring service by mid-to-high-latitude satellites. However, the limited availability of beam slots per cycle for satellites leads to significant residual traffic demand in mid-to-high-latitude cells. The GA-BH scheme focuses on maximizing the sum of demand satisfaction rate across all cells, effectively enhancing this metric by prioritizing low-demand cells at high latitudes, which results in greater residual traffic demand in mid-to-low-latitude cells. In contrast, the proposed beam-hopping scheme ensures that the residual traffic demand across all cells remain below 500 Mbps under varying energy efficiency weight factors, markedly outperforming other beam-hopping schemes. Moreover, when the energy efficiency weight factor is smaller, 95% of the traffic demand across all cells is effectively satisfied; even with a larger weight factor, cells with residual traffic demand are distributed more uniformly, effectively ensuring fairness among cells. The above results confirm that the proposed scheme achieves tangible performance gains over existing state-of-the-art beam-hopping

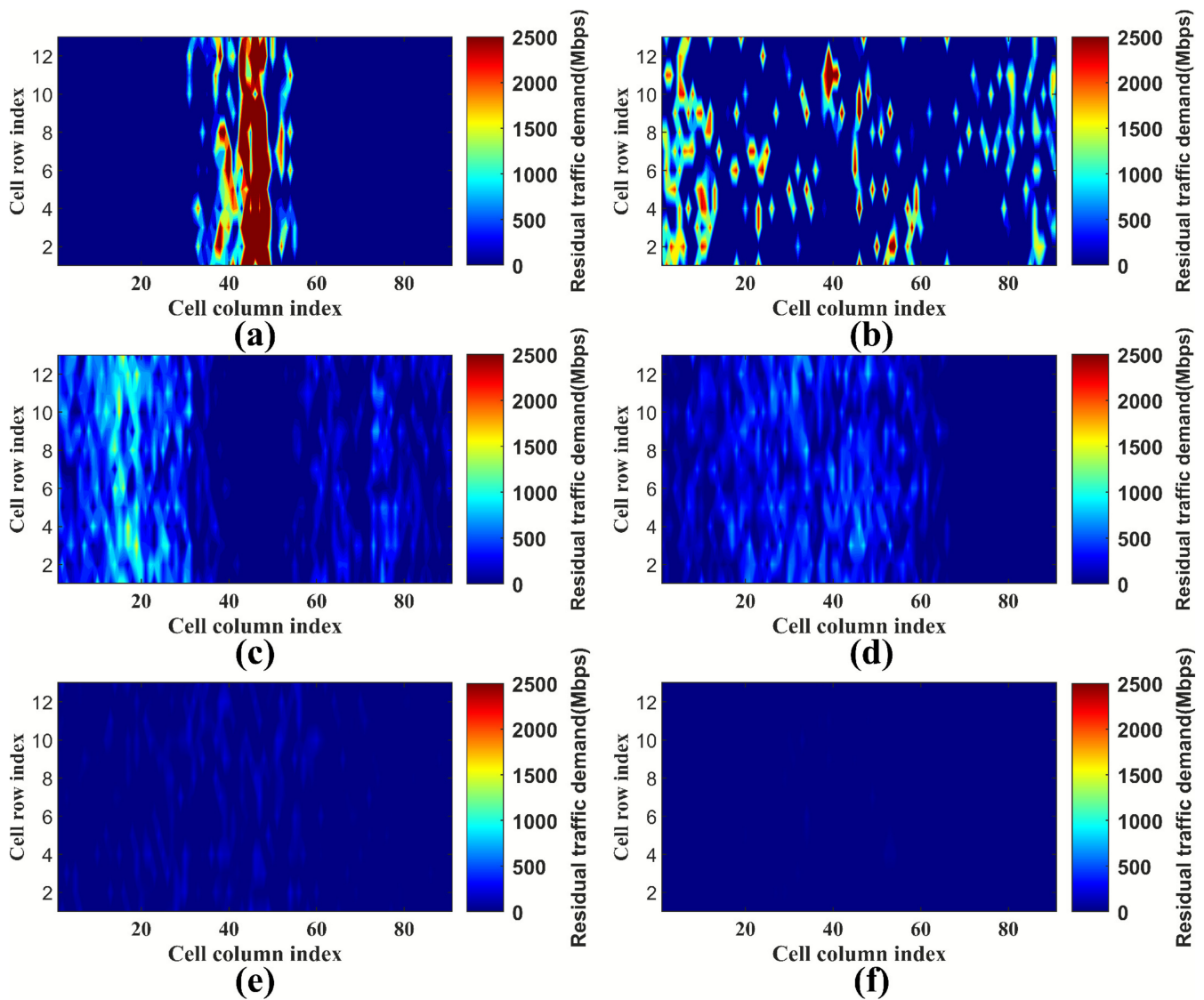


FIGURE 9 | The residual traffic demand across all cells for different beam-hopping schemes. (a) The SINR Max-BH, (b) the G-BH, (c) the JIMS-BH, (d) the GA-BH, (e) the proposed beam hopping $w = 0.7$, and (f) the proposed beam hopping $w = 0.3$.

algorithms while maintaining practical deployability through its centralized coordination framework, demonstrating clear benefits for real-world LEO–GSO coexistence systems.

5 | Conclusions

In this paper, we study the spectrum compatibility issue between LEO and GSO satellite systems, aiming to improve the RE of LEO systems without causing harmful interference to GSO users. We propose an interference avoidance technique based on adjacent satellites collaborative beam hopping to solve this problem. The residual demand first adjacent satellites collaborative greedy user scheduling strategy and the convex optimal power allocation scheme based on quadratic transformation in this technology can effectively ensure that the interference experienced by the GSO system remains below the ITU limit and enhance the system RE and capacity of LEO system. In addition, adaptive demand satisfaction ratio user grouping can adapt the degree of RE preference over energy efficiency. The simulation

findings verify that our proposed interference avoidance technology can successfully coordinate the cofrequency interference between LEO and GSO satellite systems, which offers a fresh perspective for frequency compatible sharing between two satellite systems.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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